

GearDesign Report

Cylindrical gear(pair)

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1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	18.3100	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	25.0000	53.0000
Facewidth (mm)	b	40.0000	40.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	82.1600	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Reference diameter (mm)	d	52.6660	111.6530
Tip diameter (mm)	d_a	56.666 (56.666 / 56.566)	115.653 (115.653 / 115.553)
Root diameter (mm)	d_f	47.666 (47.474 / 47.364)	106.653 (106.461 / 106.351)
Number of teeth spanned	k	4.0000	7.0000
Base tangent length (no backlash) (mm)	W_k	21.477 (21.411 / 21.374)	40.099 (40.033 / 39.996)
Effective Diameter of ball/pin (mm)	D_M	3.5000	3.5000
Diametral measurement over two balls (mm)	M_{dk}	57.533 (57.367 / 57.271)	116.636 (116.459 / 116.357)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.1052		1.1309		1.0000
Bending S.F	2.2480		2.3462		1.4000
Scuffing S.F (Flash Temp.)	2.5619				2.0000
Micropitting S.F	0.7714				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	3.8000	0.0000	1.8000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.4995
Overlap Ratio	2.0000
Total Contact Ratio	3.4995
Worst Pitch Line Velocity [m/s]	27.5761
Total Efficiency [%]	98.9818
Worst Mesh Loss [W]	688.5069
Worst Churning Loss [W]	838.8006
Worst Peak-to-Peak T.E. [μm]	0.2558
Gear Total Mass [kg]	1.2460
Hunting tooth	Complete

1.5. Summary - Load cases

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2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	18.3100	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	25.0000	53.0000
Facewidth (mm)	b	40.0000	40.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	82.1597	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h^*_aP	1.0000	1.0000
Dedendum coefficient	h^*_fP	1.2500	1.2500
Root radius coefficient	ρ^*_fP	0.3800	0.3800
Tip form height coefficient	h^*_{FaP}	0.9500	0.9500
Ramp angle (deg)	α_{KP}	50.0000	50.0000
Protuberance height coefficient	h^*_{prP}	0.0000	0.0000
Protuberance angle (deg)	α_{prP}	0.0000	0.0000
Tip alteration (mm)	k_{m_n}	0.0000	0.0000
Tip radius coefficient (for topping)	ρ^*_{aP}	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Young's modulus (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m ³)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10 ⁻⁸ /°C)	α_{coeff}	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm ²)	σ_B	1200.0000	1200.0000
Yield strength (N/mm ²)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_{ZF}	20.0000	20.0000
Mean roughness height, flank (um)	R_{ZH}	4.8000	4.8000
Roughness average value, root (um)	R_{AF}	3.0000	3.0000
Roughness average value, flank (um)	R_{AH}	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm ²)	σ_{Flim}	430.0000	430.0000
Fatigue strength for contact stress (N/mm ²)	σ_{Hlim}	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.1200	
Transverse module (mm)	m_t	2.1067	
Working normal pressure angle (deg)	α_{wn}	20.0000	
Transverse pressure angle (deg)	α_t	20.9759	
Working transverse pressure angle (deg)	α_{wt}	20.9759	
Working transverse pressure angle, e/i (deg)	$\alpha_{wt}, e/i$	20.9958 / 20.9558	
Base helix angle (deg)	β_b	17.1703	
Helix angle at operating pitch circle (deg)	β_w	18.3100	
Radial backlash without tolerance (mm)	j_{r0}	0.0000	
Circumferential backlash without tolerance (mm)	j_{t0}	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	$j_t, e/i$	0.0737 / 0.1159	0.0737 / 0.1159
Min/max normal backlash by flank tolerance for each gear (mm)	$j_n, e/i$	0.0658 / 0.1034	0.0658 / 0.1034
Circumferential backlash, e/i (mm)	$j_{tw}, e/i$	0.1390 / 0.2402	
Normal backlash, e/i (mm)	$j_{nw}, e/i$	0.1240 / 0.2143	
Angular backlash, min (deg)	j_θ, min	0.3025	0.1427
Angular backlash, max (deg)	j_θ, max	0.5226	0.2465
Effective facewidth (mm)	b_{eff}	40.0000	
Sum of the profile shift coefficients	Sum_x	-0.0000	
Generating profile shift coefficient	$x_E, e/i$	-0.0481 / -0.0756	-0.0481 / -0.0756
Prevent undercut profile shift coefficient	x_{Emin}	-0.6873	-2.5770
Max. profile shift coeff. (min top land)	x_{Emax}	1.2835	2.1302
Reference center distance, $x_1=x_2=0$ (mm)	a_d	82.1597	
Working center distance (mm)	a_w	82.1597	
Reference diameter (mm)	d	52.6665	111.6529
Operating pitch diameter (mm)	d_w	52.6665	111.6529
Operating pitch diameter with tolerance (mm)	$d_w, e/i$	52.6735 / 52.6594	111.6679 / 111.6380
27		56.6665	115.6529
Tip diameter with tolerance (mm)	$d_a, e/i$	56.6665 / 56.5665	115.6529 / 115.5529

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diamter (mm)	d_f	47.6665	106.6529
Root diameter with tolerance (mm)	d_f, e/i	47.4741 / 47.3642	106.4606 / 106.3507
Tip form diameter (mm)	d_Fa	56.4665	115.4529
Tip form diameter with tolerance (mm)	d_Fa, e/i	56.2519 / 56.1298	115.2489 / 115.1325
Root form diamter (mm)	d_Ff	49.7724	108.1574
Root form diameter with tolerance (mm)	d_Ff, e/i	49.6923 / 49.6491	108.0156 / 107.9357
Active tip diameter (EAP) (mm)	d_Na	56.4665	115.4529
Active tip diameter with tolerance (mm)	d_Na, e/i	56.2519 / 56.1298	115.2489 / 115.1325
Active root diameter (SAP) (mm)	d_Nf	50.0329	108.7853
Active root diameter with tolerance (mm)	d_Nf, e/i	50.1888 / 50.1111	109.0028 / 108.8938
Mean diameter for mass calculation (mm)	d_m	52.0178	111.0043
Gear mass (kg)	m	0.3580	0.8880
Gear inertia (kg.m^2)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.5000	0.5000
Effective tip clearance, e/i (mm)	c, e/i	0.7121 / 0.5852	0.7121 / 0.5852
Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm)	c_Nff	0.2094	0.4391

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	4.5000	4.5000
Tooth depth, e/i (mm)	h, e/i	4.6511 / 4.5462	4.6511 / 4.5462
Working depth (mm)	h_w	4.0000	
Addendum (mm)	h_a	2.0000	2.0000
Addendum, e/i (mm)	h_a, e/i	2.0000 / 1.9500	2.0000 / 1.9500
Dedendum (mm)	h_f	2.5000	2.5000
Dedendum, e/i (mm)	h_f, e/i	2.5962 / 2.6511	2.5962 / 2.6511
Number of virtual gear teeth	z_n	28.8473	61.1562
Tooth thickness coeff.	s_nc	1.5708	1.5708
Transverse tooth thickness (mm)	s_t	3.3091	3.3091
Normal tooth thickness (mm)	s_n	3.1416	3.1416
Normal tooth thickness at d_a (mm)	s_an	1.4709	1.5748
Normal tooth thickness at d_a, e/i (mm)	s_an, e/i	1.4477 / 1.3535	1.5467 / 1.4613
Normal tooth thickness at d_Fa (mm)	s_Fan	1.5733	1.6628
Normal tooth thickness at d_Fa, e/i (mm)	s_Fan, e/i	1.6672 / 1.5643	1.7297 / 1.6383
Normal space width at d_f (mm)	e_fn	0.0000	1.5877
Normal space width at d_f, e/i (mm)	e_fn, e/i	0.0000 / 0.0000	1.6134 / 1.6288

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p _n	6.2832	
Transverse pitch (mm)	p _t	6.6183	
Normal base pitch (mm)	p _{bn}	5.9043	
Transverse base pitch (mm)	p _{bt}	6.1797	
Axial pitch (mm)	p _x	20.0001	
Lead (mm)	p _z	500.0018	1060.0037

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g _α	9.2665	
Length of path of contact, e/i (mm)	g _α , e/i	8.8393 / 8.5148	
Length of approach POC of pinion (A to C) (mm)	g _{f1}	4.8173	
Length of approach POC of pinion, e/i (mm)	g _{f1} , e/i	4.5998 / 4.5581	
Length of recess POC of pinion (C to E) (mm)	g _{a1}	4.4492	
Length of recess POC of pinion, e/i (mm)	g _{a1} , e/i	4.2394 / 4.2197	
Length T1 - A (mm)	T1A	4.6093	
T1A tolerance (mm)	e/m/i	5.0152 / 4.9160 / 4.8169	
Length T1 - B (mm)	T1B	7.6961	
T1B tolerance (mm)	e/m/i	7.4765 / 7.4134 / 7.3503	
Length T1 - C (mm)	T1C	9.4266	
T1C tolerance (mm)	e/m/i	9.4365 / 9.4266 / 9.4168	
Length T1 - D (mm)	T1D	10.7890	
T1D tolerance (mm)	e/m/i	11.1948 / 11.0957 / 10.9966	
Length T1 - E (mm)	T1E	13.8758	
T1E tolerance (mm)	e/m/i	13.6562 / 13.5931 / 13.5300	
Length T2 - A (mm)	T2A	24.8018	
T2A tolerance (mm)	e/m/i	24.5634 / 24.4950 / 24.4266	
Length T2 - B (mm)	T2B	21.7150	
T2B tolerance (mm)	e/m/i	22.0915 / 21.9976 / 21.9038	
Length T2 - C (mm)	T2C	19.9844	
T2C tolerance (mm)	e/m/i	20.0053 / 19.9844 / 19.9636	
Length T2 - D (mm)	T2D	18.6221	
T2D tolerance (mm)	e/m/i	18.3837 / 18.3153 / 18.2469	
Length T2 - E (mm)	T2E	15.5353	
T2E tolerance (mm)	e/m/i	15.9118 / 15.8180 / 15.7241	
Length T1 - T2 (mm)	T1T2	29.4111	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	29.4418 / 29.4111 / 29.3803	
Diameter of single contact point A (mm)	d_A	50.0329	115.4529
Diameter of single contact point B (mm)	d_B	51.5289	112.9381
Diameter of single contact point C (mm)	d_C	52.6665	111.6529
Diameter of single contact point D (mm)	d_D	53.7021	110.7068
Diameter of single contact point E (mm)	d_E	56.4665	108.7853
Transverse contact ratio	ϵ_{α}	1.4995	
Overlap ratio	ϵ_{β}	2.0000	
Total contact ratio	ϵ_{γ}	3.4995	
Total contact ratio with tolerance	e/m/i	3.430 / 3.404 / 3.378	
Specific sliding at Tip	ζ_a	0.4719	0.6060
Specific sliding at Root	ζ_f	-1.5381	-0.8935

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	7.9000	7.9000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	16.0000	19.0000
Profile form deviation (μm)	f_fa	8.5000	8.5000
Profile slope deviation (μm)	f_Ha	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fb	10.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.5000	9.5000
Total helix deviation (μm)	F_β	14.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	25.0000	27.0000
Runout (μm)	F_r	22.0000	24.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.0000	8.0000
Single flank composite, total (μm)	F_isT	33.0000	35.0000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	31.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	4.0000	7.0000
Base tangent length (mm)	W_k	21.4769	40.0992
Base tangent length with tolerance (mm)	W_k, e/i	21.4112 / 21.3736	40.0334 / 39.9958
Tolerance of W_k (mm)	ΔW_k , e/i	-0.0658 / -0.1034	-0.0658 / -0.1034
Diameter of contact point (mm)	d_M	53.2547	111.0427
Theoretical diameter of ball/pin (mm)	D_M.theory	3.4121	3.3746
Effective Diameter of ball/pin (mm)	D_M.eff	3.5000	3.5000
Radial single ball measurement (mm)	M_rk	28.8197	58.3431
Radial single ball measurement with tolerance (mm)	M_rk, e/i	28.7368 / 28.6889	58.2543 / 58.2032
Diametral two ball measurement (mm)	M_dk	57.5326	116.6364
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	57.3671 / 57.2714	116.4589 / 116.3568
Tolerance of M_dk (mm)	ΔM_{dk}	-0.1655 / -0.2612	-0.1775 / -0.2797

3. Detail rating

3.1. General

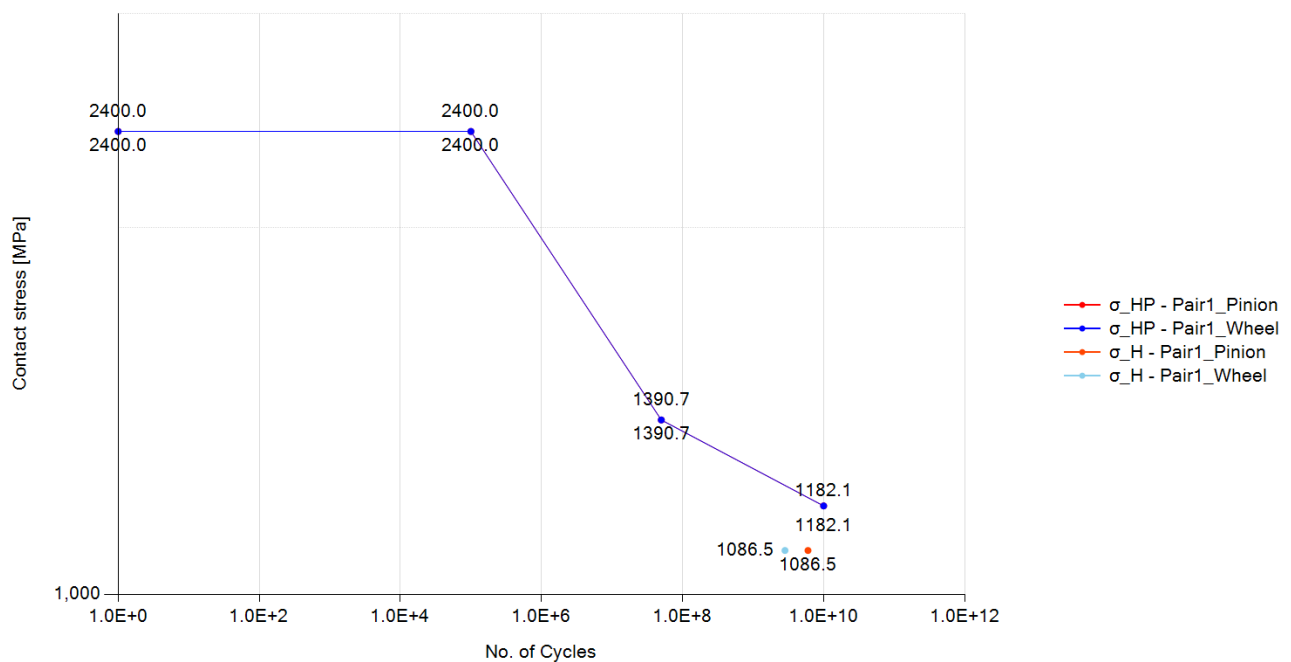
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4716.9810
Nominal tangential load at d (N)	F_t	5439.4930	
Nominal axial load at d (N)	F_a	1799.9940	
Nominal radial load at d (N)	F_r	2085.3960	
Nominal normal load at d (N)	F_n	6097.2900	
Nominal tangential load per unit facewidth (N/mm)	w	135.9870	
Circumferential speed at d (m/s)	v	27.5760	
Sliding velocity at Tip dia. (m/s)	v_g	6.8570	7.4240
Nominal tangential load at d_w (N)	F_{tw}	5439.4930	
Nominal axial load at d_w (N)	F_{aw}	1799.9940	
Nominal radial load at d_w (N)	F_{rw}	2085.3960	
Circumferential speed at d_w (m/s)	v_w	27.5760	
Running-in value (μm)	y_p	0.5930	
Running-in value (μm)	y_f	0.6370	
Effective single pitch & profile deviation (μm)	f_{pb_eff}	7.3080	
Effective single pitch & profile deviation (μm)	f_{fa_eff}	7.8620	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_{st}	1.0000	
Theoretical single stiffness (N/mm/ μm)	c'_{th}	17.5940	
Single tooth stiffness (N/mm/ μm)	c'	11.7000	
Mesh stiffness (for K_v , $KH\alpha$, $KF\alpha$), (N/mm/ μm)	$c_{y\alpha}$	16.0830	
Mesh stiffness (for $KH\beta$, $KF\beta$), (N/mm/ μm)	$c_{y\beta}$	13.6700	
Mean diameter for K_v (mm)	$d_m (K_V)$	52.1660	111.1530
Moment of inertia for K_v ($\text{kg}\cdot\text{mm}^2$)	$J (K_V)$	209.6980	2959.9670
Moment of gear inertia per unit facewidth ($\text{kg}\cdot\text{mm}$)	J^*	5.2420	73.9990
Relative gear mass per unit facewidth (kg/mm)	m^*	0.0087	0.0272
Relative gear mass per unit facewidth (kg/mm)	m_{red}	0.0066	
Resonance running speed (rpm)	n_{E1}	18888.3010	
Resonance ratio	N	0.5290	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.1620	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	$K_{H\beta}$	1.3000	
Face load factor - root	$K_{F\beta}$	1.2620	
Transverse load factor - flank	$K_{H\alpha}$	1.1370	
Transverse load factor - root	$K_{F\alpha}$	1.1370	
Number of load cycles	N_L	6.0000×10^9	2.8300×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



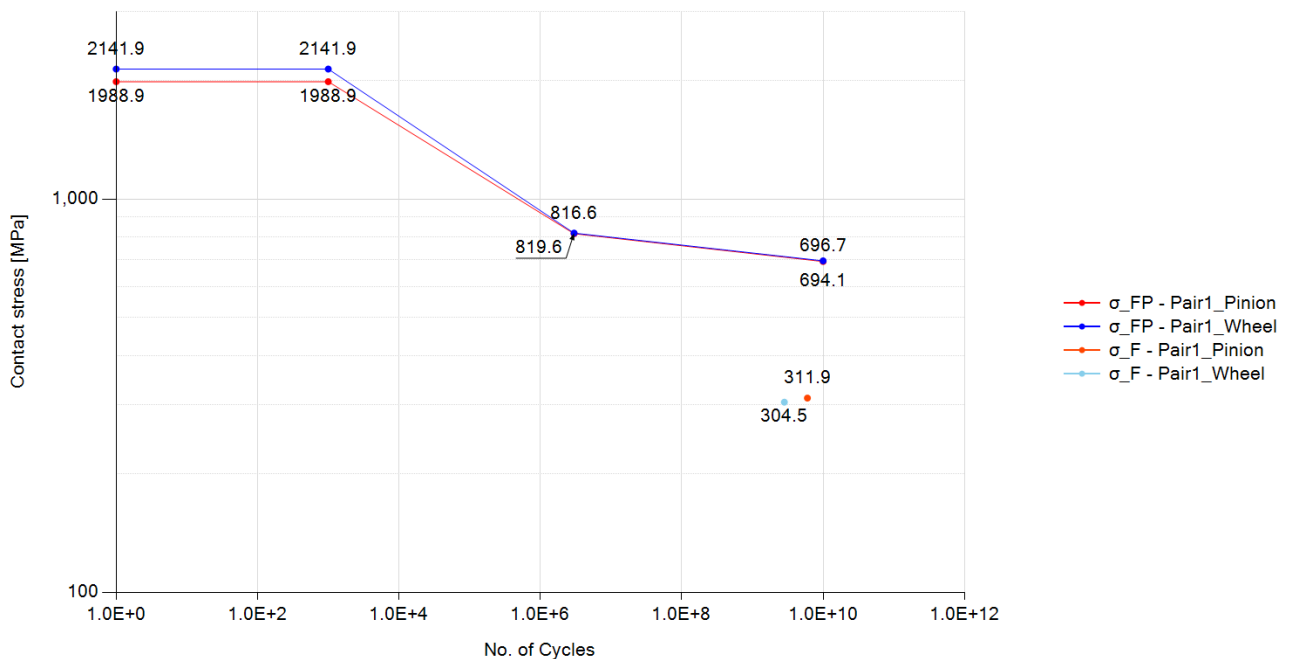
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_ϵ	0.8170	
Zone factor	Z_H	2.3910	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_β	1.0260	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm ²)	σ_{H0}	741.4800	
Contact stress (N/mm ²)	σ_H	1086.5100	1086.5100
Life factor	Z_{NT}	0.8630	0.8840

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0300	1.0300
Roughness factor	Z_R	0.9520	0.9520
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm ²)	σ_{HP_static}	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm ²)	σ_{HP_mid}	0.0000	0.0000
Permissible contact stress - reference (N/mm ²)	σ_{HP_ref}	1390.7210	1390.7210
Permissible contact stress - long (N/mm ²)	σ_{HP_long}	1182.1130	1182.1130
Permissible contact stress (N/mm ²)	σ_{HP}	1200.7810	1228.7790
Safety factor for Contact	S_H	1.1050	1.1310
Contact damage (%)	-	3.8380	1.8100
Cycle to failure	-	1.5630×10 ¹¹	1.5630×10 ¹¹

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.2150	1.0900
Stress correction factor	Y_S	1.8300	1.9910

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_{Fen}	18.8250	19.8350
Bending moment arm (mm)	h_{Fe}	2.1050	2.2150
Tooth thickness at root (mm)	s_{Fn}	4.0420	4.3650
Tooth root radius (mm)	ρ_F	1.1540	1.0370
Helix angle factor	Y_β	0.9900	
Deep tooth factor	Y_{DT}	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm ²)	σ_{F0}	149.6700	146.1240
Tooth root stress (N/mm ²)	σ_F	311.9300	304.5410
Life factor	Y_{NT}	0.8590	0.8720
Relative notch sensitive factor	$Y_{\delta relT}$	0.9920	0.9960
Surface factor	Y_{RrelT}	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_{ST}	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm ²)	σ_{FP_static}	1988.8550	2141.9300
Permissible tooth root stress - reference (N/mm ²)	σ_{FP_ref}	816.5690	819.6200
Permissible tooth root stress - long (N/mm ²)	σ_{FP_long}	694.0840	696.6770
Permissible tooth root stress (N/mm ²)	σ_{FP}	701.2240	714.5200
Safety factor for Bending	S_F	2.2480	2.3460
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	1.1060×10^{20}	4.4080×10^{20}

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_{oil}	80.0000	
Lubricant viscosity (mPa.s)	η_{oil}	13.1510	
Load stage according to FZG A/8, 3/90	S_{FZG}	12.0000	
Gear density (kg/m ³)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_{eff}	34.0000	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	291.9890	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	10.5690	
Angle factor	X_αβ	0.9700	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0480	
Maximum flash temperature (deg)	θ_fl_max	93.2960	
Mean flash temperature (deg)	θ_fl_mean	31.7140	
Total mass temperature (deg)	θ_M	97.8870	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	191.1820	
Safety factor for flash temperature	S_B	2.5620	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.0977
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.1628
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	0.7714
transverse tangential load at reference cylinder per mesh (N)	F_t	5439.4930
transverse load in plane of action (N)	F_bt	5825.5440
reduced modulus of elasticity (N/mm ²)	E_r	2.2637×10 ⁵
thermal contact coefficient of pinion (N/(ms ^{0.5} *K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms ^{0.5} *K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90°)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.2030
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm ²) ^{0.5}	Z_E	189.8120
helical load factor	K_Bγ	1.3000
load losses factor	H_v	0.1210
mean coefficient of friction	μ_m	0.0600
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.8670
buttressing factor	X_but,A	1.3000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	17.0780
sliding velocity at point A (m/s)	v_g,A	-7.4240
tangential velocity on pinion at point A (m/s)	v_r1,A	4.8270
tangential velocity on wheel at point A (m/s)	v_r2,A	12.2510
oil inlet/sump temperature (°C)	θ_oil	80.0000
bulk temperature (°C)	θ_M	103.0780
flash temperature at point A (°C)	θ_fl,A	115.7910
contact temperature at point A (°C)	θ_B,A	218.8690
pressure-viscosity coefficient at 38 °C (m ² /N)	α_38	1.8390×10 ⁻⁸
pressure-viscosity coefficient at bulk temperature (m ² /N)	α_θM	1.3110×10 ⁻⁸
pressure-viscosity coefficient at point A contact temperature (m ² /N)	α_θB,A	7.1710×10 ⁻⁹
kinematic viscosity at 40 °C (mm ² /s)	ν_40	68.0000
kinematic viscosity at 100 °C (mm ² /s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm ² /s)	ν_θB,A	1.5990
dynamic viscosity at 38 °C (Ns/m ²)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m ²)	η_θM	0.0070
dynamic viscosity at oil inlet/sump temperature (Ns/m ²)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m ²)	η_θB,A	0.0010
normal radius of relative curvature at point A (mm)	ρ_n,A	4.0680
material parameter	G_M	2968.0220
local velocity parameter at point A	U_A	6.2580×10 ⁻¹¹
local load parameter at point A	W_A	0.0003
local sliding parameter at point A	S_GF,A	0.0960
local nominal Hertzian contact stress at point A (Method B) (N/mm ²)	p_H,A	1057.4380
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm ²)	p_dyn,A	1549.4920

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.1200
Tangential speed at working pitch dia. (m/s)	v_{wt}	27.5761
Tangential force at pitch dia. (N)	F_t	5439.4932
Tooth normal force, transverse section (N)	F_{bt}	5825.5439
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_{Σ}	19.7431
Equivalent radius of curvature (mm)	ρ	6.7041
Kinematic viscosity of oil at operating temp. (cSt)	η_{oil}	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	150.0000
Mean coefficient of friction of the gear mesh	μ_{mz}	0.0378
Tooth loss factor	H_v	0.1213
Mesh power loss (W)	P_{VZP}	688.5069
Mesh efficiency (%)	η_z	99.5410

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4716.9811
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immersed depth (mm)	h_e	26.3332	55.8265
Oil density (kg/m ³)	ρ	885.0000	885.0000
Operating Viscosity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	74228.0877	74228.0877
Froude number	Fr	2943.6924	1388.5342
Centrifugal acceleration (m/s ²)	γ	14057.9491	4018.1752

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	56.6665	115.6529
Pitch diameter of gear (mm)	d_p	52.6665	111.6529
Facewidth (mm)	b	40.0000	40.0000
Dimensionless drag torque	C_m	0.0012	0.0028
Immersion area of the gear (m^2)	S_m	0.0105	0.0275
Total oil churning loss (W)	P	121.9436	716.8570
Total oil churning torque (N.m)	T	0.1164	1.4512

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000